

Field Trial of Distributed Acoustic Fiber and 6G Radio Sensing Fusion for Enhanced Localization Accuracy in an Industrial Environment

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Abstract We propose, analyse, and experimentally verify the effectiveness of combining radio and optical sensing for enhanced localization accuracy. First use cases to jointly sense within an industrial environment were demonstrated and analysed during a Field Trial at the research campus of Arena2026 in Stuttgart, Germany. ©2026 The Author(s)

Introduction

Localization of targets within industrial applications is of strong interest to improve processes and logistics within manufacturing halls. Modern industrial environments already provide deployed fiber optic systems for broadband connectivity, and indoor radio coverage via 5th generation of wireless cellular networks (5G) and beyond. Besides their primary function of data transmission, both fiber and radio technologies allow for environmental sensing and target localization [1,2].

One of the key features of 6G is integrated communication and sensing (ISAC), which allows communication hardware to sense the environment. The 3rd generation partnership project (3GPP) has identified multiple use cases for 6G ISAC, such as Automotive, Factory/Industrial, Intruder Detection, Traffic Monitoring and Unmanned Aerial Vehicles (UAV) detection [3].

Wireless ISAC solutions provide localization of targets based on reflections of radio waves in a radar-style approach [1]. Precise localization of a target depends on the system's range and angular resolution. Range resolution is related to the utilized bandwidth used for sensing and the angular resolution depends on the antenna array geometry. Conversely, in optical networks, a Distributed Acoustic Sensing (DAS) device can be used to locate mechanical vibrations (e.g. sounds, movement) near a fiber by observing the effects of the vibration on the transmitted optical signal. A target moving along (or across) a fiber can be localized using DAS if its movement is causing sufficient vibrations that can be picked up by the optical measurement equipment [4].

The limitations of each sensing system are of different natures. Communication antennas are generally designed for a trade-off between angular coverage and beam width, which results in a significant localization error in the angular domain, see Fig. 1(a) (left) [5]. In addition, the

susceptibility to partial occlusion impacts the reliability of radio-based sensing, especially in complex industrial environments where obstacles and interference are common.

On the other hand, in DAS, the sensing accuracy depends highly on the distance from the vibration source to the optical fiber, see Fig. 1(a) (right). For distances of about 5 to 10 meters and larger (depending on the deployment and the intensity of the vibration), the detectability becomes quite challenging [6].

In environments where fiber and wireless deployments cover the same area, both systems can be jointly operated to alleviate their respective limitations. Such deployments could be factory/industrial environments or road intersections, where there are both fiber deployments below ground and wireless coverage available.

In this paper, we successfully demonstrate the fusion of both radio and optical sensing methods in an industrial environment to improve localization accuracy and to mutually compensate for the limitations of both systems.

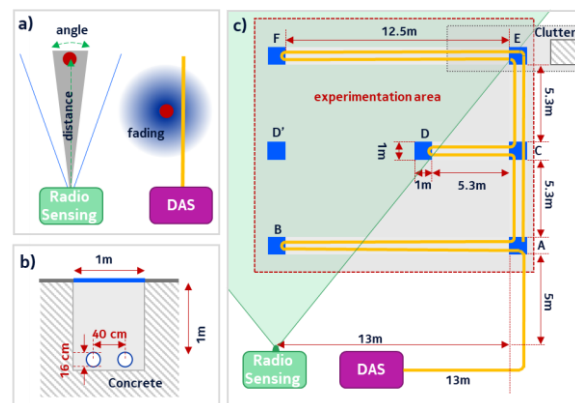


Fig. 1: (a) Limitations in sensing accuracy: Radio Sensing (left) and Optical Sensing (right), (b) Cross-section of hall floor, (c) Fiber deployment at ARENA 2036 and radio coverage of the experimentation area.

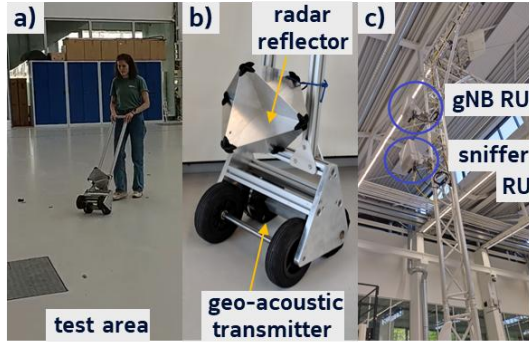


Fig. 2: (a) Photo of measurement area, (b) reference object = geo-acoustic transmitter and radar reflector, (c) sensing radio unit (RU)

Test scenario

At the Arena 2036 research campus [7], the area for our sensing tests is covered by a proof-of-concept radio sensing system. The deployment is described in Fig. 1(b,c): the optical fiber which was used for the experiment is looped through several manholes, following a path denoted as **A-B-A-C-D-C-E-F-E-C-A** and forms an “E” shape configuration. One end of the fiber cable, which is laid in ducts approximately 1 meter beneath the floor (see Fig. 1(b)), is connected to a commercial DAS device.

The DAS device is set to its smallest distance resolution, that is 0.81 m. The pulse period is set to 3 μ s, allowing for a 5 kHz acoustic bandwidth. The operating wavelength is 1550 nm. The quasi-monostatic radio sensing system [1] operates at a center frequency of 27.4 GHz using 200 MHz bandwidth and 3.5 dBm total radiated power (limited due to indoor use) using commercial 5G mmWave radio units as antenna arrays. Sensing processing is realized using the OFDM radar approach [8].

The joint sensing setup tracks the reference object pictured in Fig. 2(b): it is a trolley-like device, on which a metal octahedral radar reflector with an outer edge length of 20 cm is attached. It is moved at walking speed along the described trajectories. Note that it is purposely easy to pick-up for evaluation, which is realistic in an industrial environment, while evaluation for safety applications would require different processing.

Two test scenarios are explored, namely

Scenario 1 where the reference object is taken on a round-trip between points **E** and **F** (trajectory **F-E-F**, see points in Fig. 1(c)), and Scenario 2 where the reference object is moved at walking speed in a rectangular shape following **C-D-D'-F-E-C**. In Scenario 1, the trajectory is well covered by the DAS, while the radio sensing suffers from angular resolution around **E**. Scenario 2 is more challenging due to limitations of both systems. The segment **D-D'-F** shows no DAS coverage, and the area around **E** and **C** is either outside of the wireless coverage for radio sensing or targets are harder to detect due to clutter proximity (see Fig. 1(c)).

Real-time sensing and fusion

The DAS measurement is performed along the deployed fibers during operation of the acoustic generator. First, we obtain the vibration data in a time-distance plane, where the distance is the fiber distance, see Fig. 3(a). Because the fiber is deployed in roundtrips along the path, the trace exhibits some symmetry (45 s into the measurement, the DAS measured vibrations both at 80 m and 110 m from the fiber start). The distance axis is modified and turned into an X-Y plane to fit the actual shape of the deployed fiber: we get a two-dimensional (2D) visualization of the amount of vibration captured by the optical fiber, that changes in time as displayed in Fig. 3(b,c). Here, the coordinates are such that the origin (0,0) is set as the location of the Radio Unit (RU), see Fig. 1(c), therefore point **A** is (13.5,5). The vibration is further normalized with regards to the maximum measured vibration intensity on the measurement duration.

From a DAS native time resolution of 0.1 ms, we extract timestamps every 3 ms. For each timestamp, we identify the peak in the x-y plane representation and use it as the target's estimated position. For some timestamps, the peak magnitude can be weaker (target too far from top of the fiber, change in the pipes diameter or in the concrete density...), such as in Fig. 3(b) compared to Fig. 3(c). If the peak is too weak (according to a predefined threshold), it is considered unreliable, in which case we decide to ignore the point of some specific timestamps. Fig. 3(d)

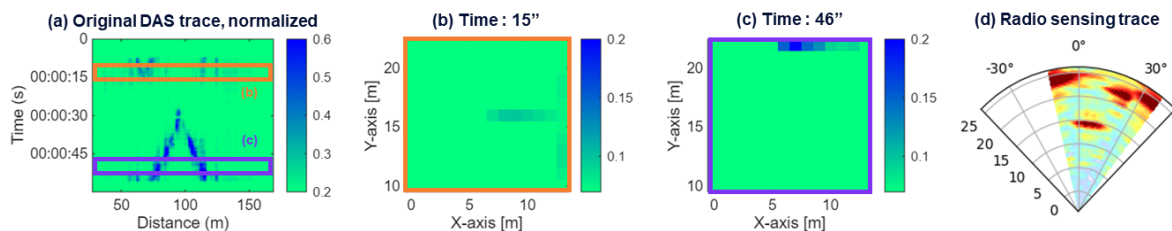


Fig. 3: (a) original DAS trace (waterfall) and (b,c) its x-y mapping for two different time instants (namely 14 s and 45 s), scenario 2. (d) Snapshot of radio sensing.

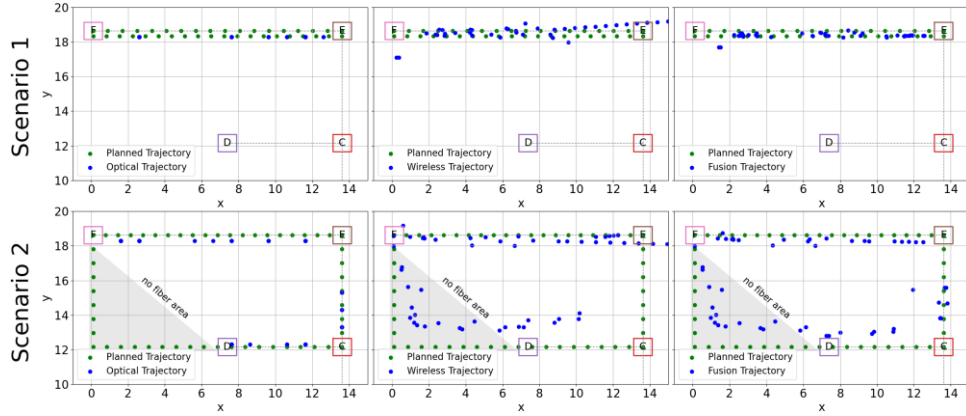


Fig. 4: Scenarios 1 (top) and 2 (bottom): (left) optical trajectory, (middle) wireless trajectory, and (right) fusion.

shows the snapshot of radio sensing at one time instant, in polar coordinates.

For the wireless system, we take the transmitted and received signals at the RUs and compute a channel state information (CSI) representation over range, Doppler, azimuth and elevation. Targets are then identified as points in this CSI exceeding a magnitude threshold. Their estimated x-y coordinates (origin at RUs) are then derived by converting the CSI's range and azimuth data from polar to Cartesian coordinates. Target position estimates are not guaranteed for every timestamp due to potential radio system's coverage gaps, e.g., around position **C** and **E**. Thus, to bridge these trajectory gaps, we used a Kalman filter [9] that provides a smooth and estimated continuation of the target's motion.

Finally, the estimated positions of the target using both systems are merged. We choose timestamps from the slowest sampling device, e.g., the radio device in this experiment. For each timestamp, we get a pair of estimated positions, where some positions can be left empty (e.g., weak DAS signal). The resulting estimated position is a weighted average of both positions divided into three cases: (i) 0.5 for each system when both are present, (ii) 1.0 when only one system is present, and (iii) 0.9 for optical and 0.1 for wireless in areas with poor wireless coverage (e.g., around **C** and **E**).

Experimental results

The results of estimated positions for Scenarios 1 and 2 are displayed in Fig. 4: optical estimated trajectory and wireless estimated trajectory are

compared to the planned trajectory, and the right plot shows the fusion trajectory. Furthermore, Table 1 gives the estimation error statistics for all configurations.

In Scenario 1, despite having a fiber path all the way, the vibrations near point **F** were too weak and therefore left empty. In wireless, the estimated trajectory was not recorded as linear due to imprecisions in the target's range and azimuth estimates and points near **E** were left empty due to radio coverage gaps. The fusion allows to retrieve a homogeneous trajectory between **F** and **E**. In Scenario 2, a large part of the trajectory occurs in the absence of fiber below the target (grey area), such that only the **D-C-E-F** part of the trajectory was recorded. The wireless sensing system suffers the same as in the 1st Scenario, namely the lack of radio coverage near the **C-E** segment. Fusion of both sets of estimates allows to retrieve the rectangle trajectory.

Conclusions

We proposed a combination of radio and fiber-optic sensing in an indoor industrial configuration. We discussed the limitations of each system and highlighted their complementarity as we proposed a combination of their estimates. Fusion of wireless and optics sensing demonstrated positioning error reduction of at least 28.5% (optics) and 26.7% (wireless) for mean error.

The fusion of distributed acoustic fiber and radio sensing paves the way for enhanced localization services, as both technologies will complement each other and will increase accuracy and reliability significantly, for indoor configuration and beyond.

Tab. 1: Results: comparison of trajectory accuracy for distributed optical sensing, radio sensing, and their combination.

	Scenario 1			Scenario 2		
	Optical	Wireless	Fusion	Optical	Wireless	Fusion
Mean error [m]	1.139	1.213	0.814	1.999	1.58	1.157
90-percentile [m]	2.058	1.869	1.436	4.708	2.67	1.727
Maximum error [m]	2.527	5.544	2.146	6.317	5.817	3.059

Acknowledgements

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